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Magnetic Material & Magnetization

by Tony Shing

Overview:

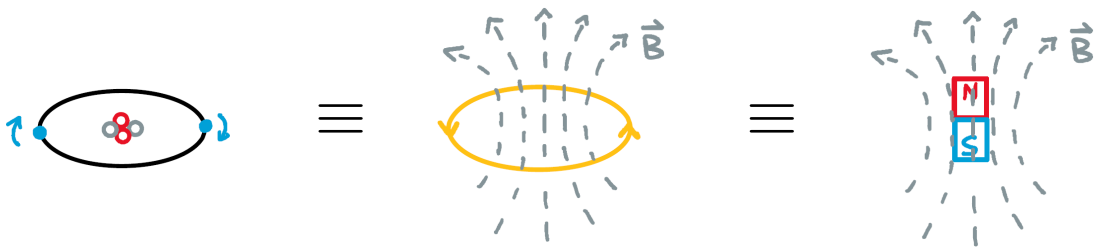
- Magnetic dipole: Origin and basic model
- How to describe dipole arrangement: Magnetization field $\vec{\boldsymbol{M}}$, bound currents $\vec{\boldsymbol{J}}_b$, $\vec{\boldsymbol{K}}_b$
- Material under external $\vec{\boldsymbol{B}}$ field:
  - A special case of material model: Linear magnetic material
  - Describe external field: Free currents $\vec{\boldsymbol{J}}_f$, $\vec{\boldsymbol{K}}_f$, auxiliary field $\vec{\boldsymbol{H}}$

1 Magnetic Dipole

1.1 Electron as Magnetic Source

Atoms's interactions with B-field are mostly contributed by electrons. The modern explanations of electrons' magnetism requires quantum mechanics, which is obviously beyond our scope. But we can still discuss how they act as magnetic dipoles in two semi-classical pictures:

1. **Orbital motion** - Electrons orbit around nucleus, similar to a constant current loop, and generate B-field like a magnet bar.



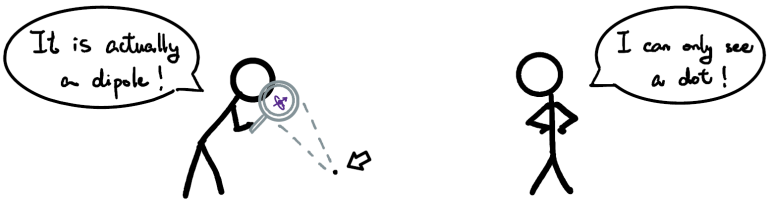
(Remember current is in opposite direction of electron flow)

2. **Spin** - Each electron carry a quantum property called spin, making them intrinsically a "point" magnetic dipole. Spin cannot be described completely by any classical picture.

Spin is usually drawn as a rotating charged arrow, although electron is a *point* particle.

Although being a "vector-like" property, quantum mechanics experiments showed that spin can only be measured one component at a time. So we say that electron spin can only have two possible values: same direction (spin up) or opposite direction (spin down) relative to a reference direction.

Since electrons are the main source of B-field and all atoms contain electrons, theoretically all materials are magnetic. However the dipoles in real life are so small ($\sim 10^{-10}$ m) and are usually packed in random orientations, the net field can be treated as zero if we view the dipoles from a very large distance.

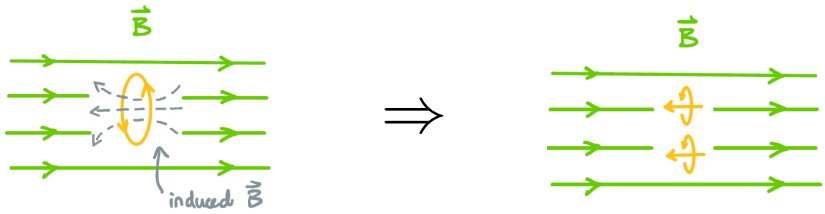


So magnetization of big objects are normally unobservable, unless we apply an external B-field to align the dipoles in some direction.

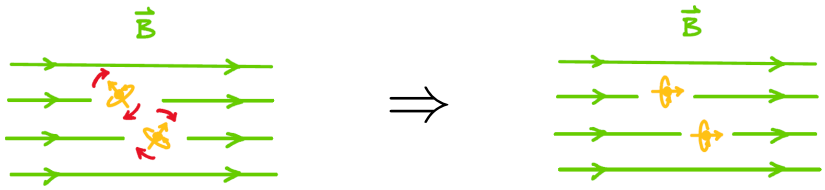
1.2 Diamagnetism V.S. Paramagnetism

Electrons' orbital motion and spin react to external B-field in the opposite ways. The true explanation requires quantum mechanics, although we may try to describe with classical pictures:

- Orbital motion are like large current loops - induced B-field is generated to oppose any external B-field, which is equivalent to creating dipoles in the opposite direction.



- Electron are points objects - they can flip their spins easily to align with B-field.

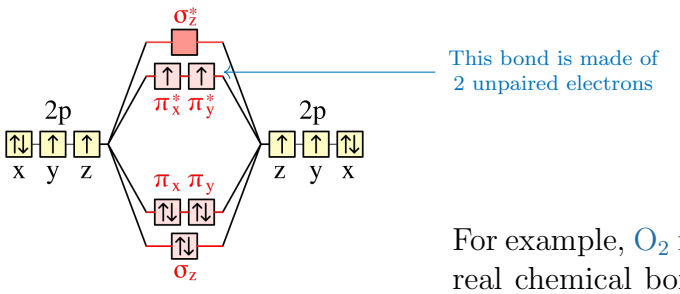


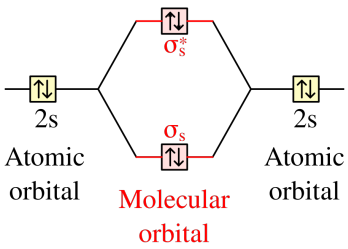
The two reponses are given the formal names:

Response	**Diamagnetism**	**Paramagnetism**
Contributed by	Orbital Motion	Spin
Response	Dipoles align opposite to applied B-field	Dipoles align the same direction as applied B-field
Relative Magnitude	Weaker	Much stronger
Found in	ALL materials	Materials that contain unpaired electrons due to unusual chemical bondings

You may have learnt the **Pauli exclusion principle** from high school chemistry, which says that electrons are stable only if they form a "spin up + spin down" pair -

- If a material does follow this principle, all spin effects are cancelled and only diamagnetism can be found. Then such material is **diamagnetic**.
- But this principle is not always true! There are materials that naturally carry unpaired electrons, such that both dia- and para-magnetism exist - and paramagnetism normally dorminates (because it is a lot stronger). Then such material is **paramagnetic**.





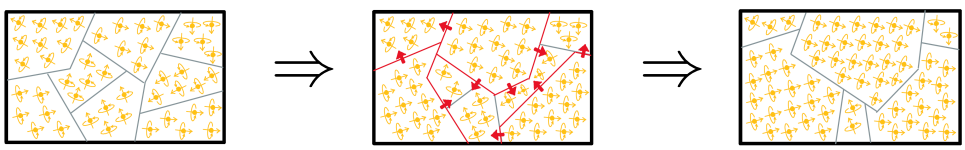
For example, O_2 molecule is paramagnetic because its real chemical bonding contain 2 unpaired electrons. But explaining this requires the **molecular orbital theory**, which is only taught in university chemistry.

Side Note:

Textbooks usually describe **ferromagnetism** as the 3rd kind of magnetic response. However its origin is very different from dia-/paramagnetism:

- Dia-/paramagnetism comes from individual atoms' response to B-field. They can exist even if there is only one atom.
- Ferromagnetism is a "group" behavior of many neighbouring atoms. Its theory requires statistical mechanics too!

On ferromagnetic materials, atoms tend to align to the same direction as their neighbours. Atoms of the same dipole direction divide the material into "domains" (typical size $\sim 10^{-6}$ m).

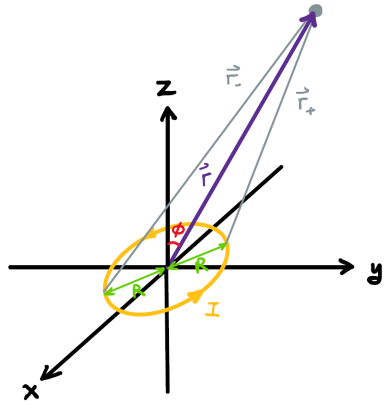


When placed under an external B-field, the domains which aligns with the field expand and "force" electrons from other domain to align with them. Finally all domains unify and form a huge magnet. This is why ferromagnetism is much stronger than the other two.

1.3 Fields of Magnetic Dipole

Since magnetic dipole is the basic unit of magnetic material, we can first analyze the field emitted by a single dipole, then sum the contributions from all dipoles according to the material structure. The classical model of magnetic dipole is as follows:

- A circular loop of radius R, carrying current I.
- We are only interested in the potential/field far away from the dipole, i.e. $|\vec{\boldsymbol{r}}| \gg R$. The position vector $\vec{\boldsymbol{r}}$ is referenced from the center of the dipole.



1.3.1 Potential by Magnetic Dipole

Deriving magnetic potential from a dipole is straightforward by Biot-Savart law, while applying Taylor approximation in the middle step.

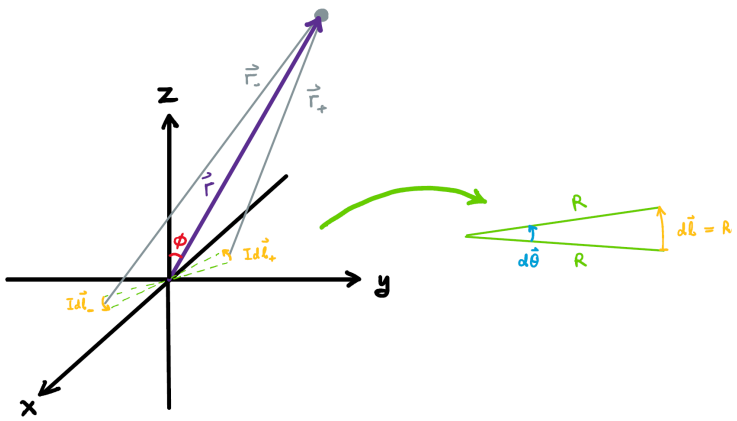
1. The total potential contributions towards position $\vec{\boldsymbol{r}}$ by two current elements at opposite position on the loop is

$$
\begin{aligned}
\mathrm{d}\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \frac{\mu_0}{4\pi}\left(\frac{I\,\mathrm{d}\vec{\boldsymbol{l}}_+}{|\vec{\boldsymbol{r}}_+|} + \frac{I\,\mathrm{d}\vec{\boldsymbol{l}}_-}{|\vec{\boldsymbol{r}}_-|}\right) \\
&= \frac{\mu_0 I}{4\pi}\left(\frac{R\,\mathrm{d}\theta}{|\vec{\boldsymbol{r}}_+|}(\hat{\boldsymbol{\theta}}) + \frac{R\,\mathrm{d}\theta}{|\vec{\boldsymbol{r}}_-|}(-\hat{\boldsymbol{\theta}})\right) \\
&= \frac{\mu_0 I R\,\mathrm{d}\theta}{4\pi}\left(\frac{1}{\sqrt{|\vec{\boldsymbol{r}}|^2 + R^2 - 2|\vec{\boldsymbol{r}}|R\cos\phi}} - \frac{1}{\sqrt{|\vec{\boldsymbol{r}}|^2 + R^2 + 2|\vec{\boldsymbol{r}}|R\cos\phi}}\right)\hat{\boldsymbol{\theta}} \\
&= \frac{\mu_0 I R\,\mathrm{d}\theta}{4\pi|\vec{\boldsymbol{r}}|}\left(\frac{1}{\sqrt{1 + \frac{R^2}{|\vec{\boldsymbol{r}}|^2} - \frac{2R}{|\vec{\boldsymbol{r}}|}\cos\phi}} - \frac{1}{\sqrt{1 + \frac{R^2}{|\vec{\boldsymbol{r}}|^2} + \frac{2R}{|\vec{\boldsymbol{r}}|}\cos\phi}}\right)\hat{\boldsymbol{\theta}}
\end{aligned}
$$

By geometry, $|\mathrm{d}\vec{\boldsymbol{l}}| = R\,\mathrm{d}\theta$

And they point in opposite direction

These are just cosine law



2. Taylor expansion "$(1+x)^n \approx 1 + nx$ for $x \ll 1$" is always used at this step. We only care about the potential at a distance much larger than the dipole's size, so take $\frac{R}{|\vec{r}|} \ll 1$,

$$\Rightarrow \frac{R^2}{|\vec{r}|^2} \approx 0$$

$$\Rightarrow \frac{1}{\sqrt{1-\frac{R}{|\vec{r}|}\cos\phi}} = \left(1 - 2\frac{R}{|\vec{r}|}\cos\phi\right)^{-\frac{1}{2}} \approx 1 + \frac{R}{|\vec{r}|}\cos\phi$$

So the potential becomes

$$\begin{aligned} \mathrm{d}\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \frac{\mu_0 I R\,\mathrm{d}\theta}{4\pi|\vec{\boldsymbol{r}}|}\left[\left(1 + \frac{R}{|\vec{\boldsymbol{r}}|}\cos\phi\right) - \left(1 - \frac{R}{|\vec{\boldsymbol{r}}|}\cos\phi\right)\right]\hat{\boldsymbol{\theta}} \\ &= \frac{\mu_0 I R\,\mathrm{d}\theta}{4\pi|\vec{\boldsymbol{r}}|} \cdot 2\frac{R}{|\vec{\boldsymbol{r}}|}\cos\phi \cdot \hat{\boldsymbol{\theta}} \end{aligned}$$

3. Before doing any integration, we first need to express every non-constant in terms of θ:
   - Recall what we learnt about polar coordinate, the angular unit vector $\hat{\boldsymbol{\theta}}_{\{r,\theta\}}$ is a function of angle θ.

$$\hat{\boldsymbol{\theta}} = -(\sin\theta)\hat{\boldsymbol{x}} + (\cos\theta)\hat{\boldsymbol{y}}$$

   - ϕ is the angle between the target's position $\vec{\boldsymbol{r}}$ and position vector of $I\,\mathrm{d}\vec{\boldsymbol{l}}_+$. Expressing them in x-y-z coordinate:
     - Target's position: $\vec{\boldsymbol{r}} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}$
     - Position vector of $I\,\mathrm{d}\vec{\boldsymbol{l}}_+$: $\vec{\boldsymbol{R}} = (R\cos\theta)\hat{\boldsymbol{x}} + (R\sin\theta)\hat{\boldsymbol{y}}$

     So that $\cos\phi$ can be calculated as their dot product:

$$\cos\phi = \frac{\vec{\boldsymbol{r}} \cdot \vec{\boldsymbol{R}}}{|\vec{\boldsymbol{r}}||\vec{\boldsymbol{R}}|} = \frac{x\cos\theta + y\sin\theta}{|\vec{\boldsymbol{r}}|}$$

4. Finally integrate over θ from 0 to π to find the total contribution of the loop.

$$\begin{aligned} \int \mathrm{d}\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \int_0^\pi \frac{\mu_0 I R\,\mathrm{d}\theta}{4\pi|\vec{\boldsymbol{r}}|} \cdot 2\frac{R}{|\vec{\boldsymbol{r}}|}\cos\phi \cdot \hat{\boldsymbol{\theta}} \\ &= \frac{\mu_0 I R^2}{2\pi|\vec{\boldsymbol{r}}|^2}\int_0^\pi \left(\frac{x\cos\theta + y\sin\theta}{|\vec{\boldsymbol{r}}|}\right)(-(\sin\theta)\hat{\boldsymbol{x}} + (\cos\theta)\hat{\boldsymbol{y}})\,\mathrm{d}\theta \\ &= \frac{\mu_0 I R^2}{2\pi|\vec{\boldsymbol{r}}|^3}\int_0^\pi (-x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}})\underbrace{\sin\theta\cos\theta}_{\int_0^\pi \sin\theta\cos\theta\,\mathrm{d}\theta = 0} + (x\hat{\boldsymbol{y}})\underbrace{\cos^2\theta}_{\int_0^\pi \cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{2}} - (y\hat{\boldsymbol{x}})\underbrace{\sin^2\theta}_{\int_0^\pi \sin^2\theta\,\mathrm{d}\theta = \frac{\pi}{2}}\,\mathrm{d}\theta \\ &= \frac{\mu_0 I R^2}{2\pi|\vec{\boldsymbol{r}}|^3} \cdot \frac{\pi}{2} \cdot (x\hat{\boldsymbol{y}} - y\hat{\boldsymbol{x}}) \end{aligned}$$

It happens that the term in the bracket is the cross product of $\hat{\boldsymbol{z}}$ and $\vec{\boldsymbol{r}}$, since

$$\hat{\boldsymbol{z}} \times \vec{\boldsymbol{r}} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ 0 & 0 & 1 \\ x & y & z \end{vmatrix} = -y\hat{\boldsymbol{x}} + x\hat{\boldsymbol{y}}$$

So that the potential becomes

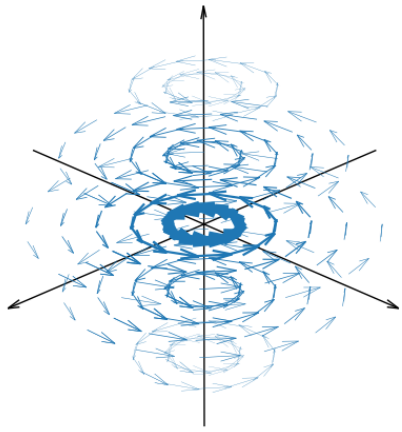
$$\begin{aligned}\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \frac{\mu_0 I R^2}{4|\vec{\boldsymbol{r}}|^2}\left(\frac{x\hat{\boldsymbol{y}} - y\hat{\boldsymbol{x}}}{|\vec{\boldsymbol{r}}|}\right) \\ &= \frac{\mu_0 I R^2}{4|\vec{\boldsymbol{r}}|^2}\frac{\hat{\boldsymbol{z}} \times \vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|} \\ &= \frac{\mu_0}{4\pi}\frac{I\pi R^2}{|\vec{\boldsymbol{r}}|^2}(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{r}})\end{aligned}$$

5. To emphasize that we now treat a magnetic dipole as "one unit of source", We define the **magnetic dipole moment $\vec{\boldsymbol{m}}$**,

$$\boxed{\vec{\boldsymbol{m}} \overset{\text{def}}{=} I \cdot \begin{pmatrix}\text{Loop's}\\ \text{Area}\end{pmatrix} \cdot \begin{pmatrix}\text{Unit vector}\\ \text{normal to loop}\end{pmatrix}}$$

In our case of a loop in x-y plane, $\vec{\boldsymbol{m}} = I(\pi R^2)\hat{\boldsymbol{z}}$. Finally we reach the standard formula of magnetic potential from a magnetic dipole:

$$\boxed{\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\frac{\vec{\boldsymbol{m}} \times \hat{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|^2}}$$



For reference, we can see its similarities with Biot-Savart law for current element:

Magnitude by I

$$\mathrm{d}\vec{\boldsymbol{A}} = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\vec{\boldsymbol{l}}}{|\vec{\boldsymbol{r}}|}$$

$\leftarrow$ Same direction as $\mathrm{d}\vec{\boldsymbol{l}}$

Proportionality constant

Decay by $\frac{1}{r}$

<u>If the source is current element</u>

Magnitude by $|\vec{\boldsymbol{m}}|$

$$\vec{\boldsymbol{A}} = \frac{\mu_0}{4\pi}\frac{\vec{\boldsymbol{m}} \times \hat{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|^2}$$

$\leftarrow$ Directional by angle between $\vec{\boldsymbol{m}}$ and $\hat{\boldsymbol{r}}$ (cross product)

Proportionality constant

Decay by $\frac{1}{r^2}$

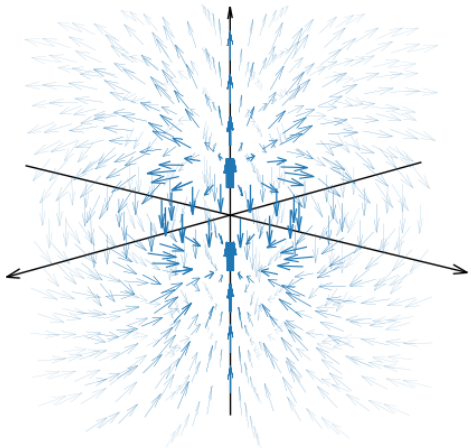
<u>If the source is dipole</u>

1.3.2 B-field from Magnetic Dipole

The standard derivation of B-field from dipole is through the relation $\vec{\boldsymbol{B}} = \vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{A}}$. It is nothing more than some boring vector calculus. Here I just quote the final result:

$$\boxed{\vec{\boldsymbol{B}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\frac{3(\vec{\boldsymbol{m}}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \vec{\boldsymbol{m}}}{|\vec{\boldsymbol{r}}|^3}}$$

(This is the same function as in electric dipole, but plotted in 3D.)



And this formula is rarely used because vector calculation is annoying.

The boring derivation of B-field from dipole:

1. Derivation is possible with high school calculus if we turn all vectors into x-y-z form: $\vec{\boldsymbol{r}} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}$, $|\vec{\boldsymbol{r}}| = \sqrt{x^2+y^2+z^2}$ and $\vec{\boldsymbol{m}} = m_x\hat{\boldsymbol{x}} + m_y\hat{\boldsymbol{y}} + m_z\hat{\boldsymbol{z}}$:

$$\begin{aligned}\frac{\vec{\boldsymbol{m}}\times\hat{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|^2} &= \frac{\vec{\boldsymbol{m}}}{|\vec{\boldsymbol{r}}|^2}\times\frac{\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|} = \frac{1}{|\vec{\boldsymbol{r}}|^3}(\vec{\boldsymbol{m}}\times\vec{\boldsymbol{r}}) = \frac{1}{|\vec{\boldsymbol{r}}|^3}\begin{vmatrix}\hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}}\\ m_x & m_y & m_z\\ x & y & z\end{vmatrix}\\ &= \frac{1}{(x^2+y^2+z^2)^{\frac{3}{2}}}[(m_yz - m_zy)\hat{\boldsymbol{x}} + (m_zx - m_xz)\hat{\boldsymbol{y}} + (m_xy - m_yx)\hat{\boldsymbol{z}}]\end{aligned}$$

2. First the x term in curl, $\left(\frac{\partial\bullet_z}{\partial y} - \frac{\partial\bullet_y}{\partial z}\right)\hat{\boldsymbol{x}}$, can be calculated by:

$$\begin{aligned}\frac{\partial\bullet_z}{\partial y} &= \frac{\partial}{\partial y}\left[\frac{(m_xy - m_yx)}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right]\\ &= \frac{m_x(x^2+y^2+z^2)^{\frac{3}{2}} - (m_xy - m_yx)\cdot 3y(x^2+y^2+z^2)^{\frac{1}{2}}}{(x^2+y^2+z^2)^3}\\ &= \frac{m_x|\vec{\boldsymbol{r}}|^2 - 3(m_xy - m_yx)y}{|\vec{\boldsymbol{r}}|^5}\end{aligned}$$

$$
\begin{aligned}
\frac{\partial \bullet_y}{\partial z} &= \frac{\partial}{\partial z}\left[\frac{(m_z x - m_x z)}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right] \\
&= \frac{-m_x(x^2+y^2+z^2)^{\frac{3}{2}} - (m_z x - m_x z)\cdot 3z(x^2+y^2+z^2)^{\frac{1}{2}}}{(x^2+y^2+z^2)^3} \\
&= \frac{-m_x|\vec{\boldsymbol{r}}|^2 - 3(m_z x - m_x z)z}{|\vec{\boldsymbol{r}}|^5}
\end{aligned}
$$

$$
\begin{aligned}
\Rightarrow \left(\frac{\partial \bullet_z}{\partial y} - \frac{\partial \bullet_y}{\partial z}\right)\hat{\boldsymbol{x}} &= \frac{2m_x|\vec{\boldsymbol{r}}|^2 - 3(m_x y - m_y x)y + 3(m_z x - m_x z)z}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{x}} \\
&= \frac{2m_x|\vec{\boldsymbol{r}}|^2 - 3m_x y^2 + 3m_y xy + 3m_z xz - 3m_x z^2 + 3m_x x^2 - 3m_x x^2}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{x}} \\
&= \frac{2m_x|\vec{\boldsymbol{r}}|^2 - 3m_x(x^2+y^2+z^2) + 3m_x x^2 + 3m_y xy + 3m_z xz}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{x}} \\
&= \frac{-m_x|\vec{\boldsymbol{r}}|^2 + 3(m_x x + m_y y + m_z z)x}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{x}} \\
&= \frac{-m_x|\vec{\boldsymbol{r}}|^2 + 3(\vec{\boldsymbol{m}}\cdot\vec{\boldsymbol{r}})x}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{x}}
\end{aligned}
$$

3. Similar for y and z terms:

$$
\begin{aligned}
\left(\frac{\partial \bullet_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} &= \frac{-m_y|\vec{\boldsymbol{r}}|^2 + 3(\vec{\boldsymbol{m}}\cdot\vec{\boldsymbol{r}})y}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{y}} \\
\left(\frac{\partial \bullet_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}} &= \frac{-m_z|\vec{\boldsymbol{r}}|^2 + 3(\vec{\boldsymbol{m}}\cdot\vec{\boldsymbol{r}})z}{|\vec{\boldsymbol{r}}|^5}\hat{\boldsymbol{z}}
\end{aligned}
$$

4. Summing all 3 components gives:

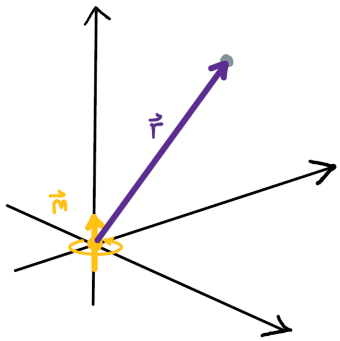
$$
\begin{aligned}
\vec{\boldsymbol{\nabla}}\times\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}} \\
&= \frac{\mu_0}{4\pi}\frac{3(\vec{\boldsymbol{m}}\cdot\vec{\boldsymbol{r}})(x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}) - (m_x\hat{\boldsymbol{x}} + m_y\hat{\boldsymbol{y}} + m_z\hat{\boldsymbol{z}})|\vec{\boldsymbol{r}}|^2}{|\vec{\boldsymbol{r}}|^5} \\
&= \frac{\mu_0}{4\pi}\frac{3(\vec{\boldsymbol{m}}\cdot\vec{\boldsymbol{r}})\vec{\boldsymbol{r}} - \vec{\boldsymbol{m}}|\vec{\boldsymbol{r}}|^2}{|\vec{\boldsymbol{r}}|^5} \\
&\qquad\qquad \text{Take out magnitude to become unit vector} \\
&= \frac{\mu_0}{4\pi}\frac{3(\vec{\boldsymbol{m}}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}}\,|\vec{\boldsymbol{r}}|^2 - \vec{\boldsymbol{m}}|\vec{\boldsymbol{r}}|^2}{|\vec{\boldsymbol{r}}|^5}
\end{aligned}
$$

$$
\boxed{\vec{\boldsymbol{B}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\frac{3(\vec{\boldsymbol{m}}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \vec{\boldsymbol{m}}}{|\vec{\boldsymbol{r}}|^3}}
$$

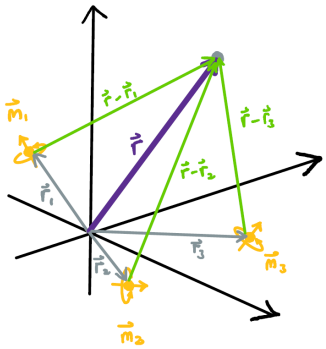
2 Describing Dipole Arrangement

In general, all materials are made of many tiny magnetic dipoles. To analyze the material's magnetic properties, we can begin with the property of a single dipole, then sum the contributions of all dipoles according to the dipole arrangement in the material.

$$\underbrace{\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\frac{\vec{\boldsymbol{m}}_i}{|\vec{\boldsymbol{r}}|^2} \times \left(\frac{\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|}\right)}_{\text{By a single dipole from origin}} \qquad \underbrace{\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\sum_{\text{All dipoles } i}\frac{\vec{\boldsymbol{m}}_i}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i|^2} \times \left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}_i|}\right)}_{\text{By many dipoles at different positions } \vec{\boldsymbol{r}}_i}$$



$\Rightarrow$



Now we introduce two quantities that can be used to describe magnetic dipole arrangements.

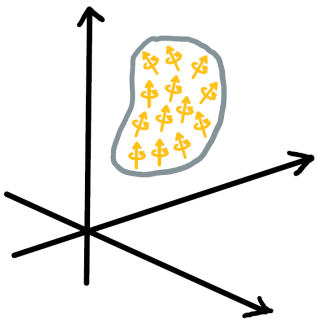
- **Magnetization field** - $\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}})$
- **Bound current distributions** - $\vec{\boldsymbol{J}}_b(\vec{\boldsymbol{r}})$ and $\vec{\boldsymbol{K}}_b(\vec{\boldsymbol{r}})$

2.1 Magnetization Field

When the dipoles in the material are so dense such that we can treat them as a continuous distribution, we can replace

- Sum of all dipoles $\xRightarrow{\text{become}}$ Volume integral over the whole object.
- Discrete dipoles source $\xRightarrow{\text{become}}$ A vector distribution called **magnetization field** $\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}})$.

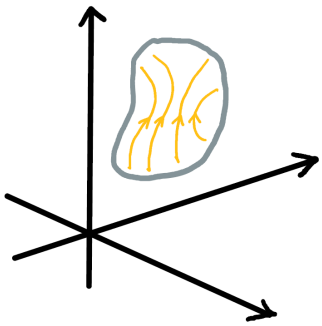
$$\boxed{\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\iiint_{\substack{\text{Whole}\\\text{material}}}\frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2} \times \left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)\mathrm{d}^3\vec{\boldsymbol{r}}'}$$



$\Rightarrow$ Connecting the dipoles resemble a vector field

$\rightarrow$ call it "$\vec{\boldsymbol{M}}$ field"

$\Rightarrow$



The magnetization field $\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}})$ can also be interpreted as **magnetic dipole density** because its usage is similar to charge density $\vec{\boldsymbol{J}}(\vec{\boldsymbol{r}})$. Comparing with the Biot-Savart law for magnetic potential:

- When the source is made of current elements:

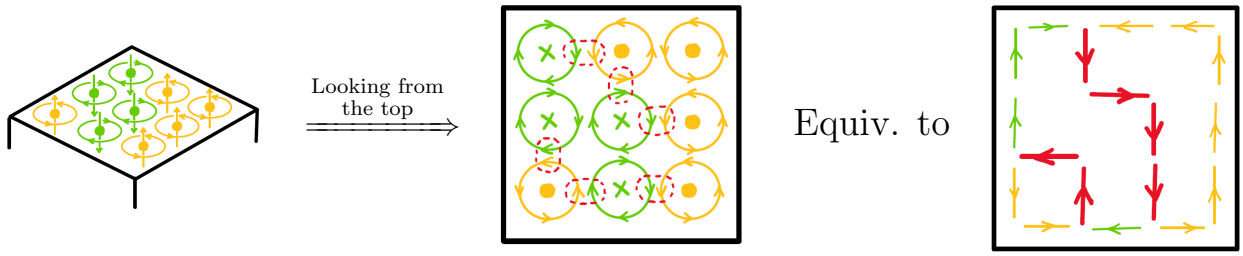
$$\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\iiint \frac{\vec{\boldsymbol{J}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\,\mathrm{d}^3\vec{\boldsymbol{r}}' \quad\sim\quad \frac{\mu_0}{4\pi}\sum\frac{(\text{current density})}{(\text{distance})}$$

- When the source is made of dipoles:

$$\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\iiint \frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\times\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)\mathrm{d}^3\vec{\boldsymbol{r}}' \quad\sim\quad \frac{\mu_0}{4\pi}\sum\frac{(\text{dipole density})}{(\text{distance})^2}\times\left(\substack{\text{unit}\\\text{vector}}\right)$$

2.2 Bound Current Distribution

Ultimately, magnetic dipoles are just loops of currents. If the dipoles are aligned non-uniformly in a material, some regions may appear to have a higher flow of current in one direction than the other way.



These flows are described as **bound current** - they are always "bounded" to regions where the current flow is not cancelled. With vector calculus, we can derive their relation with $\vec{\boldsymbol{M}}$:

$$\begin{aligned}
\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \frac{\mu_0}{4\pi}\iiint_{\substack{\text{Whole}\\\text{material}}} \frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\times\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)\mathrm{d}^3\vec{\boldsymbol{r}}' \\
&= (\cdots\ \textit{After more boring vector calculus}\ \cdots) \\
&= \frac{\mu_0}{4\pi}\iiint_{\substack{\text{Whole}\\\text{material}}} \frac{\vec{\boldsymbol{\nabla}}\times\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\,\mathrm{d}^3\vec{\boldsymbol{r}}' + \frac{\mu_0}{4\pi}\oiint_{\substack{\text{Surface}\\\text{of material}}} \frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\times\mathrm{d}^2\vec{\boldsymbol{r}}'
\end{aligned}$$

Notice that these two integrals are very similar to Biot-Savat law of magnetic vector potential:

$$\begin{cases}
\dfrac{\mu_0}{4\pi}\displaystyle\iiint_{\substack{\text{Whole}\\\text{material}}} \frac{\vec{\boldsymbol{\nabla}}\times\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\,\mathrm{d}^3\vec{\boldsymbol{r}}' \quad\sim\quad \dfrac{\mu_0}{4\pi}\displaystyle\sum_{\substack{\text{Inside}\\\text{material}}} \frac{(\vec{\boldsymbol{\nabla}}\times\vec{\boldsymbol{M}})}{(\text{distance})} \\[2ex]
\dfrac{\mu_0}{4\pi}\displaystyle\oiint_{\substack{\text{Surface}\\\text{of material}}} \frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\times\mathrm{d}^2\vec{\boldsymbol{r}}' \quad\sim\quad \dfrac{\mu_0}{4\pi}\displaystyle\sum_{\substack{\text{On material}\\\text{surface}}} \frac{\left(\substack{\text{Surface component}\\\text{of }\vec{\boldsymbol{M}}}\right)}{(\text{distance})}
\end{cases}$$

What if we interpret these two as currents?

Their compositions suggest that we can define two new current densities expressions:

- **Volume bound current density**

$$\boxed{\vec{\boldsymbol{J}}_b(\vec{\boldsymbol{r}}) \overset{\text{def}}{=} \vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{M}}(\vec{\boldsymbol{r}})}$$

- **Surface bound current density**

$$\boxed{\vec{\boldsymbol{K}}_b(\vec{\boldsymbol{r}}) \overset{\text{def}}{=} \vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}) \times \hat{\boldsymbol{n}}}$$

  Here the $\hat{\boldsymbol{n}}$ is the (outward) unit normal vector on the material surface. $\vec{\boldsymbol{M}} \times \hat{\boldsymbol{n}}$ means we we are taking the component perpendicular to normal vector, i.e. a component parallel to the surface.

Then the potential formula can be interpreted as:

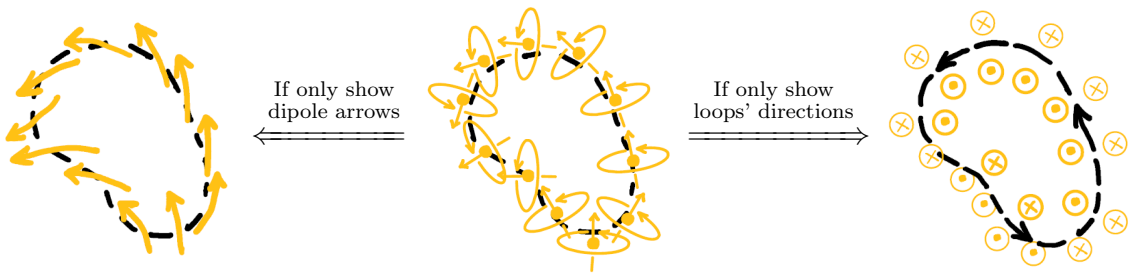
$$\begin{aligned}\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) &= \frac{\mu_0}{4\pi} \underset{\substack{\text{Whole}\\\text{material}}}{\iiint} \frac{\vec{\boldsymbol{J}}_b(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}'|} \mathrm{d}^3\vec{\boldsymbol{r}}' + \frac{\mu_0}{4\pi} \underset{\substack{\text{Surface}\\\text{of material}}}{\oiint} \frac{\vec{\boldsymbol{K}}_b(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}'|} \mathrm{d}^2\vec{\boldsymbol{r}}' \\ &= \begin{pmatrix}\text{Contribution}\\\text{by currents inside}\\\text{the material}\end{pmatrix} + \begin{pmatrix}\text{Contribution}\\\text{by currents on}\\\text{material's surface}\end{pmatrix}\end{aligned}$$

which is simply saying that the magnetic potential from a magnetic material is contributed by two distributions of currents - one flows inside the material and one only flows on the surface.

2.2.1 Volume Bound Current Density

Consider drawing an Ampere's loop and check how many dipoles are pointing in the same direction as we travel along the loop,

More dipole align with the circle	$\Leftrightarrow$	Stronger current at the center



Recall what we have learnt about Ampere's law:

- We can use sign of dot product line integral to identify positions with vector field in rotation pattern.
- Curl operator $\vec{\boldsymbol{\nabla}}\times$ is equivalent to finding dot product line integral per area.

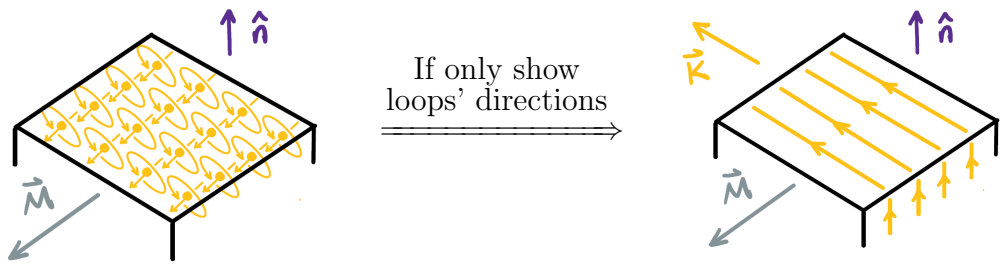
This is how we can interpret the definition:

$$\begin{pmatrix}\text{Bound current}\\\text{volume density}\end{pmatrix} \sim \vec{\boldsymbol{J}}_b = \vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{M}} \sim \begin{pmatrix}\text{\# of dipoles per area}\\\text{around a rotation center}\end{pmatrix}$$

2.2.2 Surface Bound Current Density

Consider where the $\vec{\boldsymbol{M}}$ field wraps around the material surface,

More dipole align in the same direction	$\Leftrightarrow$	More current flow on the material surface



Since we choose the normal vector $\hat{\boldsymbol{n}}$ to be pointing outward of the material's surface, direction of bound current flow is exactly in the direction of $\vec{\boldsymbol{M}} \times \hat{\boldsymbol{n}}$.

$$\begin{pmatrix}\text{Bound current}\\ \text{surface density}\end{pmatrix} \sim \vec{\boldsymbol{K}}_b = \vec{\boldsymbol{M}} \times \hat{\boldsymbol{n}} \sim \begin{pmatrix}\text{Same direction dipole}\\ \text{per surface area}\end{pmatrix}$$

The boring derivation:

1. First we need to show that the unit vector part is related to gradient:

Note: Differentiation is w.r.t. x', y', z'

$$\begin{aligned}
\vec{\boldsymbol{\nabla}}\left(\frac{1}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right) &= \left(\hat{\boldsymbol{x}}\frac{\partial}{\partial \underline{x'}} + \hat{\boldsymbol{y}}\frac{\partial}{\partial \underline{y'}} + \hat{\boldsymbol{z}}\frac{\partial}{\partial \underline{z'}}\right)\left(\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\right) \\
&= \frac{\hat{\boldsymbol{x}}(x-x') + \hat{\boldsymbol{y}}(y-y') + \hat{\boldsymbol{z}}(z-z')}{[(x-x')^2+(y-y')^2+(z-z')^2]^{\frac{3}{2}}} \\
&= \frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^3} \\
&= \frac{1}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2}\underbrace{\left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)}
\end{aligned}$$

This part is just a unit vector in $\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'$ direction

This is basically $\frac{\mathrm{d}}{\mathrm{d}r'}\left(\frac{1}{r-r'}\right) = \frac{1}{(r-r')^2}$ but in vector calculus version. An additional unit vector is present because result of gradient must be a vector field.

2. Substitute into the potential formula to become:

$$\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\iiint \underbrace{\frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|^2} \times \left(\frac{\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)}_{\text{Result from step 1}} \mathrm{d}^3\vec{\boldsymbol{r}}'$$

$$= \frac{\mu_0}{4\pi}\iiint \underbrace{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}') \times \vec{\boldsymbol{\nabla}}\left(\frac{1}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)}_{\vec{\boldsymbol{G}}\times\vec{\boldsymbol{\nabla}}f \quad (\text{Note the order of } f \text{ and } \vec{\boldsymbol{G}})} \mathrm{d}^3\vec{\boldsymbol{r}}'$$

$$= \frac{\mu_0}{4\pi}\iiint \underbrace{-\vec{\boldsymbol{\nabla}} \times \left(\frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right)}_{-\vec{\boldsymbol{\nabla}}\times(f\vec{\boldsymbol{G}})} + \underbrace{\frac{\vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}}_{f\vec{\boldsymbol{\nabla}}\times\vec{\boldsymbol{G}}} \mathrm{d}^3\vec{\boldsymbol{r}}'$$

Here we used the **product rule of curl** from line 2 to line 3:

$$\boxed{\vec{\boldsymbol{\nabla}} \times (f\vec{\boldsymbol{G}}) = f\vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{G}} + (\vec{\boldsymbol{\nabla}}f) \times \vec{\boldsymbol{G}}}$$

where f is a scalar function (like $\frac{1}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}$) and $\vec{\boldsymbol{G}}$ is a vector function (like $\vec{\boldsymbol{M}}$).

3. Finally use the "divergence theorem for curl" to convert the 1st term's volume integral into a surface integral over the volume's surface:

$$\vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) = \frac{\mu_0}{4\pi}\underbrace{\iiint \vec{\boldsymbol{\nabla}} \times \left(\frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|}\right) \mathrm{d}^3\vec{\boldsymbol{r}}'} + \frac{\mu_0}{4\pi}\iiint \frac{\vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|} \mathrm{d}^3\vec{\boldsymbol{r}}'$$

Divergence theorem for curl

$$= \frac{\mu_0}{4\pi}\underbrace{\oiint \frac{\vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|} \times \mathrm{d}^2\vec{\boldsymbol{r}}'} + \frac{\mu_0}{4\pi}\iiint \frac{\vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{M}}(\vec{\boldsymbol{r}}')}{|\vec{\boldsymbol{r}}-\vec{\boldsymbol{r}}'|} \mathrm{d}^3\vec{\boldsymbol{r}}'$$

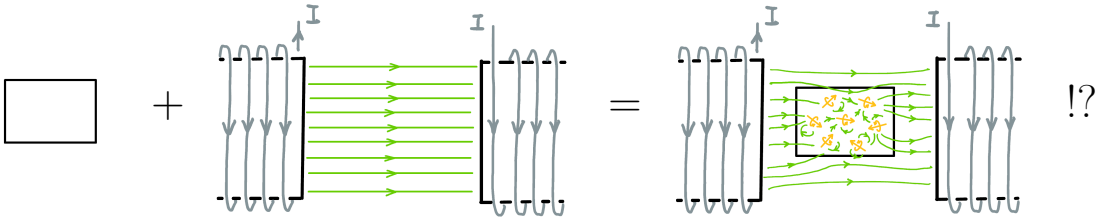
$$= \begin{pmatrix}\text{Contribution}\\ \text{from material's}\\ \text{surface}\end{pmatrix} + \begin{pmatrix}\text{Contribution}\\ \text{from inside}\\ \text{the material}\end{pmatrix}$$

3 Material under External B-field

Recall that magnetic response of atoms is the total contribution by their electrons' orbital motion and spin, which exhibit completely opposite behaviours under and external B-field.

Paramagnetic dipoles Diamagnetic dipoles

However in a material structure, the dipole density from the two behaviours in the material can vary by position. So the final dipole alignment might not always align with the external B-field!



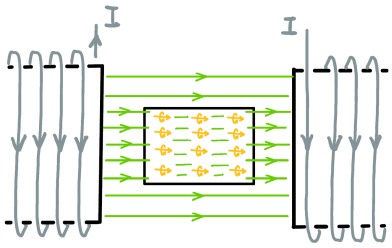
In reality, fully model a material's response to B-field can be very complicated. The general theory assumes the magnetization field $\vec{\boldsymbol{M}}$ (i.e. new dipole alignment) as some function to the total B-field $\vec{B}_{\text{total}}$,

$$\vec{\boldsymbol{M}} \;=\; f(\vec{\boldsymbol{B}}_{\text{total}}) \;\sim\; \underbrace{\vec{\boldsymbol{a}}_i^{(1)} B_i + \vec{\boldsymbol{a}}_{ij}^{(2)} B_i B_j + \vec{\boldsymbol{a}}_{ijk}^{(3)} B_i B_j B_k + \cdots}_{\text{Like a Taylor expansion}}$$

This f function denotes a theoretical model that we have choosen to investigate the material. And the model parameters $\vec{\boldsymbol{a}}^{(1)}, \vec{\boldsymbol{a}}^{(2)}, \vec{\boldsymbol{a}}^{(3)}$... shall be determined from experiment.

3.1 Special Case: Linear Magnetic Material

Linear magnetic material is the simplest model of B-field response - where the dipoles are completely free to rotate, such that new dipole alignment is directly proportional to the external B-field's magnitude and direction. This assumption is applicable to most daily life materials.



No random arrangement.
Every dipole is well-aligned.

The linear model only has 1 parameter to be determined from experiments - the proportionality constant between $\vec{\boldsymbol{M}}$ and $\vec{\boldsymbol{B}}_{\text{total}}$.

$$\vec{\boldsymbol{M}} = \left(\substack{\text{Some}\\ \text{constant}}\right) \cdot \vec{\boldsymbol{B}}_{\text{total}}$$

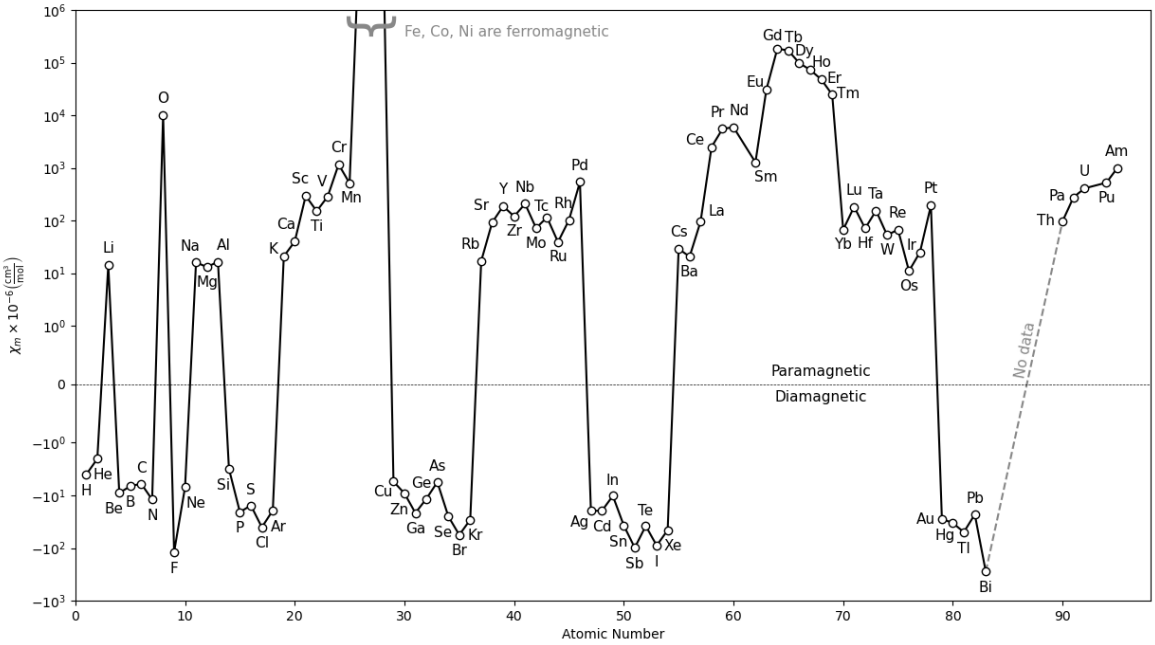
Unlike using $\vec{\boldsymbol{P}}$ v.s. $\vec{\boldsymbol{E}}$ in linear dielectric model, linear magnetization model is almost always written by $\vec{\boldsymbol{M}}$ v.s. a new kind of field $\vec{\boldsymbol{H}}$. We shall explain what $\vec{\boldsymbol{H}}$ is right after.

$$\boxed{\vec{\boldsymbol{M}} = \chi_m \vec{\boldsymbol{H}}}$$

Here the χ_m is called **magnetic susceptibility**, a pure number (no unit) whose value depends on the type of material.

- $\chi_m = 0$ for vacuum (by defintion) - there are no dipoles in a vacuum.
- $\chi_m < 0$ for diamagnetism - since their dipoles are opposite to the B-field's direction.
- $\chi_m > 0$ for paramagnetism - since their dipoles follow the B-field's direction.

Note: Ferromagnetism generally requires non-linear models, so they do not have constant χ_m.



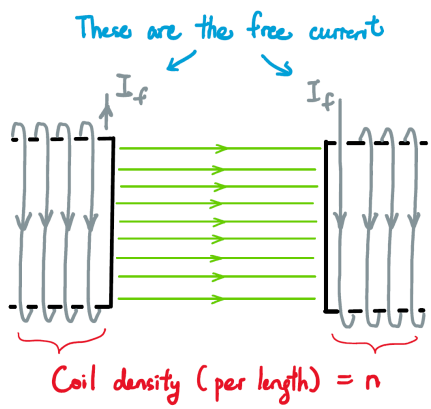
χ_m by elements. Data from *CRC Handbook of Chemistry and Physics*.

3.2 Describing External Field

The total B-field is a result of the external B-field plus the field induced by dipole alignment. After we have chosen the model about dipole alignment, now we look at the external B-field.

3.2.1 Free Current Distribution

In order to create an external B-field, we need a "setup" to build an external source of currents. For example, apply current through two solenoids:



The currents that flow in the setup are given the name **free currents**, to distinguish from the bound currents (dipole alignment) in the material. The are "free" because we can always control them by varying the setup, making them known quantities in calculation.

The actual thing we can control = Current

$$\mu_0 n I_{\text{free}} = |\vec{\boldsymbol{B}}|$$

B-field depends on applied current

For calculation, free current densities are denoted like bound currents densities:

- **Volume free current density** $\vec{\boldsymbol{J}}_f(\vec{\boldsymbol{r}})$ - Current distribution inside the setup.
- **Surface free current density** $\vec{\boldsymbol{K}}_f(\vec{\boldsymbol{r}})$ - Current on any surfaces of the setup.

3.2.2 Auxiliary Field $\vec{\boldsymbol{H}}$

Previously, we have relate bound currents to a vector field quantity - the magnetization field $\vec{\boldsymbol{M}}$. Similarly, free current can be related with another vector field. From Ampere's law,

$$\begin{aligned}
\frac{1}{\mu_0}\vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{B}}_{\text{total}} &= \vec{\boldsymbol{J}}_{\text{total}} = \begin{pmatrix}\text{All}\\ \text{currents}\end{pmatrix} \\
&= \vec{\boldsymbol{J}}_f + \vec{\boldsymbol{J}}_b = \begin{pmatrix}\text{Free currents}\\ \text{on setup}\end{pmatrix} + \begin{pmatrix}\text{Bound currents}\\ \text{on material}\end{pmatrix} \\
&= \vec{\boldsymbol{J}}_f + \vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{M}} \\
\vec{\boldsymbol{\nabla}} \times \left(\frac{1}{\mu_0}\vec{\boldsymbol{B}} - \vec{\boldsymbol{M}}\right) &= \vec{\boldsymbol{J}}_f
\end{aligned}$$

Here we define the **Auxiliary field** $\vec{\boldsymbol{H}}$:

$$\boxed{\vec{\boldsymbol{H}} \overset{\text{def}}{=} \frac{1}{\mu_0}\vec{\boldsymbol{B}} - \vec{\boldsymbol{M}}} \longleftarrow \text{This relation connects all 3 field quantities}$$

Such that it is related to the free current density by:

$$\boxed{\begin{cases} \vec{\boldsymbol{J}}_f(\vec{\boldsymbol{r}}) \overset{\text{def}}{=} \vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{H}}(\vec{\boldsymbol{r}}) \\ \vec{\boldsymbol{K}}_f(\vec{\boldsymbol{r}}) \overset{\text{def}}{=} \vec{\boldsymbol{H}}(\vec{\boldsymbol{r}}) \times \hat{\boldsymbol{n}} \end{cases}}$$

where $\hat{\boldsymbol{n}}$ is the (outward) unit normal vector on the equipment surface. Cross product with it to represent a direction parallel to the surface.

Here the relations of $\vec{\boldsymbol{H}}$ with free currents are almost identical to $\vec{\boldsymbol{B}}$ with total currents in Ampere's law. In the special case of linear magnetic material, we can solve for $\vec{\boldsymbol{H}}$ from the given free currents like the Ampere's law integral form or with Biot-Savart law.

$$\begin{aligned}
I_f &= \oint \vec{\boldsymbol{H}}(\vec{\boldsymbol{r}}) \cdot \mathrm{d}\vec{\boldsymbol{l}} \\
&= \oint \underline{|\vec{\boldsymbol{H}}||\,\mathrm{d}\vec{\boldsymbol{l}}\,|\cos\theta} \longleftarrow \text{Just dot product } \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta \\
&= \underline{|\vec{\boldsymbol{H}}|}\ \underline{\cos\theta} \oint |\,\mathrm{d}\vec{\boldsymbol{l}}\,| \\
&\qquad \text{Same magnitude everywhere, Can move out of integral} \qquad \text{Form same angle everywhere, Can move out of integral} \\
&= |\vec{\boldsymbol{H}}|\ \cos\theta\ (\text{Perimeter of loop})
\end{aligned}$$

$$\boxed{|\vec{\boldsymbol{H}}| = \frac{I_f}{(\text{Perimeter of loop})\cos\theta}}$$

But note that this is incorrect for other models of magnetic material.

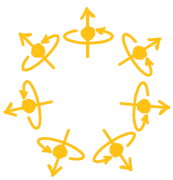
Side Note:

AlIn magnetostatics, if we know the total current distribution, we can solve for the total B-field uniquely by one PDE (Ampere's Law). However we cannot do this to $\vec{\boldsymbol{H}}$ even though it relates to free current by a similar PDE.

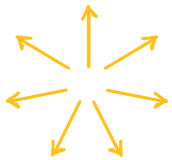
$$\boxed{\begin{array}{c}\text{Solve } \vec{\boldsymbol{\nabla}} \times \vec{\boldsymbol{B}}_{\text{total}} = \mu_0 J_{\text{total}} \\ \Downarrow \\ \vec{\boldsymbol{B}}_{\text{total}} \text{ has a unique solution}\end{array}} \quad \textbf{BUT} \quad \boxed{\begin{array}{c}\text{Solve } \vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{H}} = J_f \\ \Downarrow \\ \vec{\boldsymbol{H}} \text{ still has many possibilities}\end{array}}$$

This is because:

- $\vec{\boldsymbol{B}}_{\text{total}}$ is guarenteed to be "divergent-less", such that we can define a vector potential function $\vec{\boldsymbol{A}}$ and solve for $\vec{\boldsymbol{B}}_{\text{total}}$ uniquely.
- $\vec{\boldsymbol{M}}$ may not be "divergent-less" since the dipole arrangement can be arbituary, e.g. if they arrange into some vortex-like pattern. Then by $\vec{\boldsymbol{H}} = \frac{1}{\mu_0}\vec{\boldsymbol{B}}_{\text{total}} - \vec{\boldsymbol{M}}$, there is no guarenteed that $\vec{\boldsymbol{H}}$ is "divergent-less" either.



Equiv. to



But in the special case of linear magnetic material,

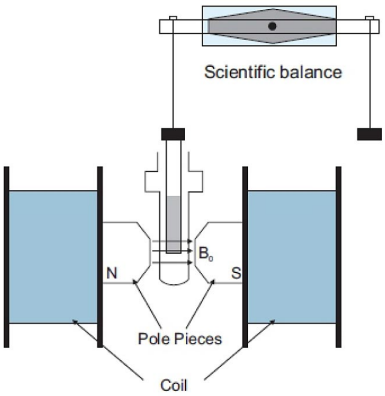
$$\boxed{\begin{array}{c}\vec{\boldsymbol{M}} \text{ aligns} \\ \text{with } \vec{\boldsymbol{B}}_{\text{total}}\end{array}} \Rightarrow \boxed{\begin{array}{c}\vec{\boldsymbol{M}} \text{ is} \\ \text{"divergent-less"}\end{array}} \Rightarrow \boxed{\begin{array}{c}\vec{\boldsymbol{H}} = \dfrac{1}{\mu_0}\vec{\boldsymbol{B}}_{\text{total}} - \vec{\boldsymbol{M}} \\ \text{is also "divergent-less"}\end{array}}$$

So it is fine to solve for $\vec{\boldsymbol{H}}$ like normal Ampere's law problems for linear magnetic material.

3.3 Measurement in Practice

In experients that involve magnetic material,

- It is impossible to measure $\vec{\boldsymbol{M}}$ - We cannot put any sensors inside a material without breaking it. The best we can do is to choose a good theoretical model that fits the material's atomic arrangement. *(E.g. Given a linear magnetic material ...)*
- $\vec{\boldsymbol{H}}$ is control-able - $\vec{\boldsymbol{H}}$ is related to free current, which can be directly controlled by a power supply with an ammeter.
- $\vec{\boldsymbol{B}}$ is measure-able - Commonly by measurinng the magnetic force exerted on the material. For example, you can take a look at the Gouy method.



The Guoy balance, a traditional setup for measuring material's response to B-field.

It is purely mechanical so measurements are not affected by EM induction.

(Image source from here)

Once we have chosen a model of the material, we can connect the two measurables by:

$$\vec{\boldsymbol{H}} = \frac{1}{\mu_0}\vec{\boldsymbol{B}}_{\text{total}} - f(\vec{\boldsymbol{B}}_{\text{total}})$$

Then the model parameters in $f(\cdots)$ can be determined by experiments. But in the special case of linear magnetic material, since there is only 1 constant parameter χ_m for each material,

$$\begin{aligned}
\vec{\boldsymbol{H}} &= \frac{1}{\mu_0}\vec{\boldsymbol{B}}_{\text{total}} - \chi_m\vec{\boldsymbol{H}} \\
(1+\chi_m)\vec{H} &= \frac{1}{\mu_0}\vec{\boldsymbol{B}}_{\text{total}} \\
\vec{\boldsymbol{H}} &= \frac{1}{\mu_0(1+\chi_m)}\vec{\boldsymbol{B}}_{\text{total}} \\
&= \frac{1}{(\text{A constant})}\cdot\vec{\boldsymbol{B}}_{\text{total}}
\end{aligned}$$

We usually use this new constant to represent the material's response to B-field, by defining

$$\boxed{\mu \overset{\text{def}}{=} \mu_0(1+\chi_m) \overset{\text{def}}{=} \mu_0\mu_r} \qquad \text{such that} \qquad \vec{\boldsymbol{H}} = \frac{1}{\mu}\vec{\boldsymbol{B}}_{\text{total}}$$

where

- μ = **Absolute permeability**, or **magnetic permeability**. Has the same unit as μ_0.
- μ_r = **Relative permeability**. A pure number. $\begin{cases} < 1 \text{ for diamagnetism} \\ > 1 \text{ for paramagnetism} \end{cases}$.

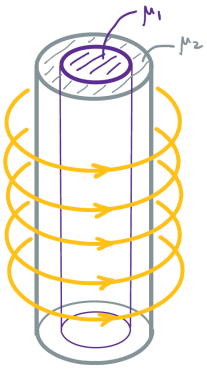
Comparing with the dielectric case where ϵ is used much more than χ_e, material scientists prefer χ_m but not μ when talking about magnetic material. This is because they wrote the theory as $\vec{\boldsymbol{M}} \propto \vec{\boldsymbol{H}}$ but not $\vec{\boldsymbol{M}} \propto \vec{\boldsymbol{B}}$, for certain reasons:

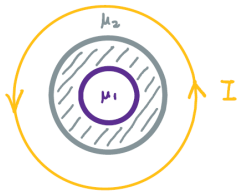
- $\chi_m = 0$ draws a clear boundary for the weak materials - whether it is diamagnetic ($\chi_m < 0$) or paramagnetic ($\chi_m > 0$). Using $\mu_r = 1 + \chi_m$ is much less intuitive.
- Historically, engineers like to call $\vec{\boldsymbol{H}}$ as the "magnetic field" - the fundamental thing that they can control in experiments. While $\vec{\boldsymbol{B}}$ is only called "magnetic flux density", as the magnetic response of material under $\vec{\boldsymbol{H}}$. (And this is surely very confusing to people who study theoretical physics.)

3.4 Calculation Example

Here are some examples of questions you may see in the chapter of linear magnetic material. In principle, one of the information about $\vec{\boldsymbol{H}}$ (free current) or $\vec{\boldsymbol{B}}_{\text{total}}$ (force) will be given, then you are asked to find the other, and so as $\vec{\boldsymbol{M}}$ and bound currents.

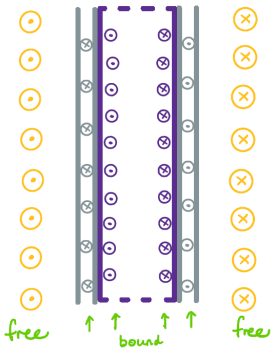
Example 3.1. Consider a coaxial cable made of a core line wrapped by some sheath material, each with absolute permeability $\mu_1 > 1$ (paramagnetic) and $\mu_2 < 1$ (diamagnetic). The cable is placed in the middle of a solenoid running with a constant current I. Assume the solenoid has a number of turn per unit length n, and is long enough such that the B-field is uniform.

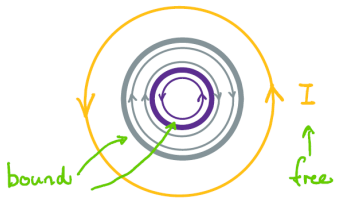




Top cross-section

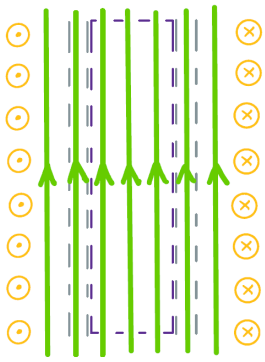
1. A good start is to label all the currents' locations, and if they are free/bounded. Note that current around a solenoid is exactly like a surface current.

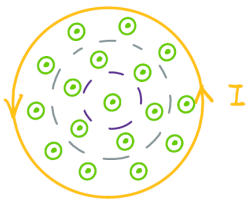




Note: Diamagnetic material ($\mu < 1$) weakens external B-field, so bounded current must be in opposite rotation. Opposite for paramagnetic materials.

2. Always remember that $\vec{\boldsymbol{H}}$ is only related to the free current. In this configuration, I is the exact amount of free current. We can treat it like the material does not exist.



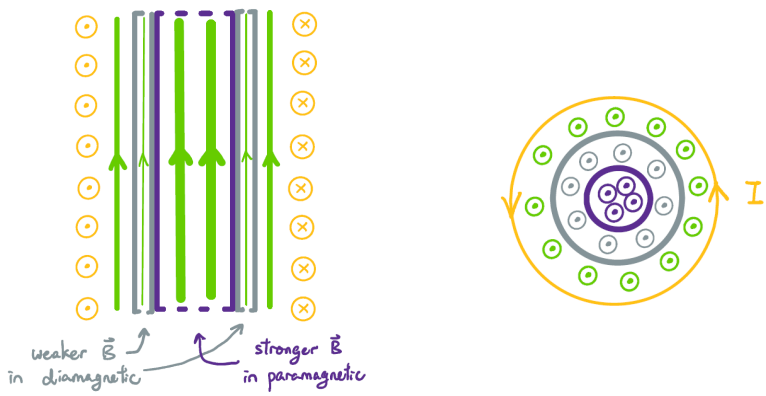


$|\vec{\boldsymbol{H}}| = nI$
is the same everywhere
inside the coils.

3. According to the linear magnetic material model, $\vec{\boldsymbol{H}}$-field is proportional to $\vec{\boldsymbol{M}}$ by the susceptibility at that location, i.e. $\vec{\boldsymbol{B}} = \mu\vec{\boldsymbol{H}}$. Because we have the same $\vec{\boldsymbol{H}}$ everywhere,

   - In vacuum, $|\vec{\boldsymbol{B}}| = \mu_0|\vec{\boldsymbol{H}}|$
   - In material 1, $|\vec{\boldsymbol{B}}| = \mu_1|\vec{\boldsymbol{H}}|$
   - In material 2, $|\vec{\boldsymbol{B}}| = \mu_2|\vec{\boldsymbol{H}}|$

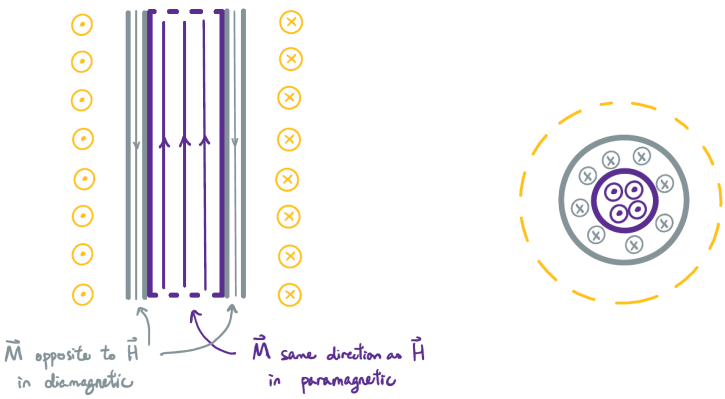
   Note: Since χ_m of most diamagnetic materials are tiny ($\sim -10^{-4}$ to -10^{-5}), μ_2 is still expected to be > 0.



4. Then we can find $\vec{\boldsymbol{M}}$ by $\vec{\boldsymbol{M}} = \dfrac{\vec{\boldsymbol{B}}}{\mu_0} - \vec{\boldsymbol{H}}$:

   - In vacuum, $|\vec{\boldsymbol{M}}| = \dfrac{|\vec{\boldsymbol{B}}|}{\mu_0} - |\vec{\boldsymbol{H}}| = \dfrac{\mu_0|\vec{\boldsymbol{H}}|}{\mu_0} - |\vec{\boldsymbol{H}}| = 0 \quad$ (Obviously)
   - In material 1, $|\vec{\boldsymbol{M}}| = \dfrac{\mu_1|\vec{\boldsymbol{H}}|}{\mu_0} - |\vec{\boldsymbol{H}}| = \left(\dfrac{\mu_1}{\mu_0} - 1\right)|\vec{\boldsymbol{H}}| \quad (= \chi_{m1}|\vec{\boldsymbol{H}}|,\ > 0)$
   - In material 2, $|\vec{\boldsymbol{M}}| = \dfrac{\mu_2|\vec{\boldsymbol{H}}|}{\mu_0} - |\vec{\boldsymbol{H}}| = \left(\dfrac{\mu_2}{\mu_0} - 1\right)|\vec{\boldsymbol{H}}| \quad (= \chi_{m2}|\vec{\boldsymbol{H}}|,\ < 0)$

   According to $\vec{\boldsymbol{M}} \times \hat{\boldsymbol{n}} = \vec{\boldsymbol{K}}_b$, the $|\vec{\boldsymbol{M}}|$ on each material are also the amount of surface bound current densities on each of the materials' surfaces.



— The End —